

AnyCam 2.5.0-rc1.0 Audit Report

First rc on the 2.5.0 line. Delivers Lorex/Dahua DVR Family Support — the consumer side of work that began as data scaffolding in 2.4.0-rc2.0 plus the Plan-2 RTSP OPTIONS Fingerprint Helper. Closes the discovery gap that left the Lorex D861A8B-Z stuck at Verifying for 90s before failing on AnyCam 2.4.x.

Already in place before this build

Two pieces of the original Plan landed earlier and are unchanged by this build: (1) the CAMERA_DB entry for the Lorex/Dahua DVR-NVR Family with full streaming_recipe data (added 2.4.0-rc2.0), plus three sibling DVR family entries also seeded with channel_iterate recipes; (2) the _rtsp_options_fingerprint helper and the brand-id rtsp_realm_regex matching support (Plan 2). Also already in place: skip_layer2 enforcement in find_rtsp_path, which prevents Layer 2 from grinding 90s on this brand.

What 2.5.0-rc1.0 adds (the consumer side)

Feature	Where	Mechanism
streaming_recipe expansion	find_rtsp_path	When brand_entry has channel_iterate recipe, expand to 34 paths (16 channels capped at 16)
Populated-channel SDP filter	_probe_rtsp_paths_single	When walker_populated_channel_test=='sdp_has_video_track', after DESCRIBE
auth_attempt_lockout policy	_probe_rtsp_paths_single	When walker_throttle_type=='auth_attempt_lockout', a Digest-auth rejection (vs 90s)
_sdp_has_video_track helper	module-level	Regex-based codec detection. 9 unit-level cases pass.
_expand_channel_iterate_paths helper	module-level	Recipe expansion with channel_cap=16. Per-channel ordering main before sdp

Explicitly deferred from this rc

Three items from the plan are NOT in 2.5.0-rc1.0 and are tracked for follow-up: (1) Multi-channel-multi-card surfacing — currently a Lorex DVR with N populated channels surfaces as ONE card showing the FIRST populated channel; surfacing all N channels as N cards requires architectural changes to the scanner's per-IP-port card output. (2) Empirical SDP shape data from a known-empty Lorex channel — user accepted shipping the populated-channel heuristic blind on the rationale that we will always be guessing for untested hardware; if field test surfaces phantom or missing cards, a 2.5.0-rc1.1 bumpfix refines the heuristic. (3) NVR channel ranges >16 — cap is hardcoded to 16 in this rc.

Validation procedure

1. Scan the network with AnyCam 2.5.0-rc1.0 with a Lorex D861A8B-Z (or any Dahua-family DVR) reachable. The DVR's IP should surface as a card titled 'Lorex / Dahua DVR-NVR Family'. 2. Scan log should show: 'RTSP path list: brand=Lorex / Dahua DVR-NVR Family streaming_recipe channel_iterate expanded to 34 paths (channels capped at 16)'. 3. Enter correct credentials. Cred-auth should complete in <10s (vs 90s on 2.4.x) and return a working stream URL pointing at the first populated channel. 4. Repeat with WRONG credentials. Cred-auth should fail in <2s with one used attempt against the camera's 10-attempt lockout counter — log shows 'auth_attempt_lockout brand — bailing walk after first auth rejection (preserves remaining attempts before camera lockout)'. 5. Hikvision .33 and Hipcam .22 behavior unchanged — different brand entries, different code paths, no recipe collision.

Pre-ship verification

verify_release.py: all 5 sections passed (compile + top-level definitions, function contracts, throttle field validation, best-practice audit, version consistent at 2.5.0-rc1.0, CHANGELOG top entry matches). Helper sanity checks: 18 unit cases across _sdp_has_video_track and _expand_channel_iterate_paths all pass.

Files changed

File	Change
camera_discovery.py	Module-level _SDP_VIDEO_CODEC_RE regex, _sdp_has_video_track helper, _expand_channel_iterate_
camera_discovery.py	find_rtsp_path: streaming_recipe expansion + host_meta plumbing for walker_streaming_recipe_active, w
camera_discovery.py	_probe_rtsp_paths_single_socket: SDP populated-channel filter (when walker_populated_channel_test='s
camera_discovery.py	_probe_rtsp_paths_single_socket: auth_attempt_lockout bail-on-first-rejection (when walker_throttle_type=
camera_discovery.py	CURRENT_VERSION → 2.5.0-rc1.0
config.yaml	version → 2.5.0-rc1.0
CHANGELOG.md	Prepended 2.5.0-rc1.0 entry